American Computer Science League

2022-2023 ● Contest 2: Binary Counting ● Senior Division

PROBLEM: Given a string of characters found on the keyboard, convert each character in the string to the binary equivalent of its ASCII code. In the resulting concatenated string, search for the increasing sequence of binary numbers starting with 0, 1, 10, 11, … until a number cannot be found anywhere in the string. Look from the start of the string. If the binary number is found, remove that occurrence of the binary number from the string. Then look from the end of the string. If the binary number is found, remove that occurrence of the binary number from the string. Once a number in the sequence cannot be found, convert the string to an octal number without leading 0s. Then repeat this same process with a sequence of base 8 numbers 0, …, 7, 10, 11, …. Output the last octal number, converted to base 10, that can successfully be found. If 0 cannot be found, output -1.

EXAMPLE: For the string "Roses are red.", convert it to a concatenated string of binary numbers using each character's ASCII code as follows:

Char	ASCII	Binary
R	82	01010010
o	111	01101111
s	115	01110011
e	101	01100101
s	115	01110011
sp	32	00100000
a	97	01100001

Char	ASCII	Binary
r	114	01110010
e	101	01100101
sp	32	00100000
r	114	01110010
e	101	01100101
d	100	01100100
.	46	00101110

Now search for binary numbers beginning with 0 in the following string:

<u>0</u>1010010 01101111 01110011 01100101 01110011 00100000 01100001
01110010 01100101 00100000 01110010 01100101 01100100 0010111<u>0</u>

Remove the 0 from both ends so the string becomes:

<u>1</u>010010 01101111 01110011 01100101 01110011 00100000 01100001
01110010 01100101 00100000 01110010 01100101 01100100 001011<u>1</u>

Remove the 1 from both ends so the string becomes:

0<u>10</u>010 01101111 01110011 01100101 01110011 00100000 01100001
01110010 01100101 00100000 01110010 01100101 01100100 00<u>10</u>11

Remove 10 from both ends so the string becomes:

0010 0<u>11</u>01111 01110011 01100101 01110011 00100000 01100001

01110010 01100101 00100000 01110010 01100101 01100100 0011

Remove 11 from both ends so the string becomes:

0010 001111 01110011 01100101 01110011 00100000 01100001
01110010 01100101 00100000 01110010 01100101 01100100 00

After removing 100 from both ends as shown above, the process continues until the final string becomes:

00001100000110000100100100000000001000100

The string 1101 cannot be found in this final string. The binary numbers 0, 1, 10, 11, 100, 101, 110, 111, 1000, 1001, 1010, 1011, and 1100 were found and deleted from one or both sides of the string, but not 1101.

The resulting string converted to an octal number is 603022200104. Repeating the same process, the sequence of strings is 603022200104, 6302220014, 630222004, 6302004, 602004, 60200 so the last octal number that can be found is 4 since 5 cannot be found in the string.

INPUT: A string containing any character that can be found on the keyboard. The string will be fewer than 200 characters.

OUTPUT: Once a number in the sequence cannot be found, output the last octal number, converted to base 10, that can be found in the string after handling all deletions as explained above. If 0 cannot be found, output -1.

SAMPLE INPUT:

```
1. Roses are red.
2. A is for Alpha; B is for Bravo; C is for Charlie.
3. A stitch in time saves nine.
4. 1, 2: Buckle my shoe! 3, 4: Shut the door!
5. The quick brown fox jumped over the lazy dogs.
```

SAMPLE OUTPUT:

```
1. 4
2. 9
3. 8
4. 6
5. 5
```

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TEST DATA

TEST INPUT:

```
1. ACSL is 45 years old and going strong.
2. What was the first computer programming language you
   learned?
3. Lions and Tigers and Bears, Oh My! This is from The Wizard
   of Oz.
4. zyxwvutsrqponmlkjihgfedcba
5. ~{w|x|y|z}
```

TEST OUTPUT:

```
1. 6
2. 9
3. 9
4. 2
5. -1
```

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问题：给定键盘上的字符串，将字符串中每个字符转换为与ASCI I 码等值的二进制数。生成的连接字符串中，搜索以 0、1、10、11…开始的递增二进制数数列，直到出现一个在字符串中任意位置都无法找到的二进制数。从字符串的开头进行查找，如果找到这个二进制数，从字符串中移除这个二进制数。然后从字符串的末尾进行查找，如果找到这个二进制数，从字符串中移除这个二进制数。一旦数列中出现一个无法被找到的数字时，将字符串转换为不以 0 开头的八进制数。然后用八进制数字组成的数列 0、…、7、10、11、…重复相同的过程。输出可以找到的最后一个八进制数的十进制等值数。如果无法找到 0，则输出 -1。

示例：对于字符串“Roses are red.” 运用如下所示每个字符的 ASCII 码，将其转换成二进制数的连接字符串。

字符	ASCII	二进制数
R	82	01010010
o	111	01101111
s	115	01110011
e	101	01100101
s	115	01110011
sp	32	00100000
a	97	01100001

字符	ASCII	二进制数
r	114	01110010
e	101	01100101
sp	32	00100000
r	114	01110010
e	101	01100101
d	100	01100100
.	46	00101110

现在搜索以下字符串中从 0 开始的二进制数：

<u>0</u>1010010 01101111 01110011 01100101 01110011 00100000 01100001
01110010 01100101 00100000 01110010 01100101 01100100 0010111<u>0</u>

从字符串的两端删除 0，使字符串变为：

<u>1</u>010010 01101111 01110011 01100101 01110011 00100000 01100001
01110010 01100101 00100000 01110010 01100101 01100100 0010111<u>1</u>

从字符串的两端删除 1，使字符串变为：

0<u>10</u>010 01101111 01110011 01100101 01110011 00100000 01100001
01110010 01100101 00100000 01110010 01100101 01100100 00<u>10</u>11

从字符串的两端删除 10，使字符串变为：

0010 0<u>11</u>01111 01110011 01100101 01110011 00100000 01100001

01110010 01100101 00100000 01110010 01100101 01100100 0011

从字符串的两端删除 11，使字符串变为：

0010 001111 01110011 01100101 01110011 00100000 01100001
01110010 01100101 00100000 01110010 01100101 01100100 00

将上方显示的 100 从两端移除后，继续这个过程直到最后的字符串变为：

0000110000011000010010010000000001000100

在最终的字符串中无法找到字符串1101。二进制数 0、1、10、11、100、101、110、111、1000、1001、1010、1011 和 1100 在字符串中找到并从字符串的一端或两端删除，但没有找到和删除 1101。

将最终生成的字符串转换为一个八进制数，即 603022200104。重复相同的过程，字符串数列为 603022200104、6302220014、630222004、6302004、602004、60200，因为在字符串中不能找到 5，所以最后一个可以找到的八进制数是 4。

输入：包含键盘上任意字符的字符串。字符串将少于 200 个字符。

输出：一旦数列中出现一个无法找到的数字，输出在完成上述所有删除操作之后的字符串中可以找到的最后一个八进制数的十进制等值数。如果无法找到 0，则输出 -1。

样本输入：

```
1. Roses are red.
2. A is for Alpha; B is for Bravo; C is for Charlie.
3. A stitch in time saves nine.
4. 1, 2: Buckle my shoe! 3, 4: Shut the door!
5. The quick brown fox jumped over the lazy dogs.
```

样本输出：

```
1. 4
2. 9
3. 8
4. 6
5. 5
```

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测试数据

测试输入：

```
1. ACSL is 45 years old and going strong.
2. What was the first computer programming language you
   learned?
3. Lions and Tigers and Bears, Oh My! This is from The Wizard
   of Oz.
4. zyxwvutsrqponmlkjihgfedcba
5. ~{w|x|y|z}
```

测试输出：

```
1. 6
2. 9
3. 9
4. 2
5. -1
```